Conversational Elicitation of Patient Utilities Using Large Language Model Simulation

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1 Executive Summary

This proposal describes a research and software-development initiative to investigate **conversational elicitation** as a method for estimating patient utilities and treatment preferences.

Traditional utility elicitation methods such as:

- standard gamble,
- time tradeoff,
- visual analog scales,
- discrete choice experiments, and
- swing weighting

often impose substantial cognitive burden on patients and may fail to capture the lived experience of disease and treatment side effects.

This project proposes an alternative approach based on:

1. open-ended conversational interviews,
2. large language model (LLM) simulation of synthetic patients,
3. inference of latent patient utility structures from narrative responses, and
4. round-trip simulation experiments evaluating recovery accuracy.

The initial disease domain will be **oral squamous cell carcinoma (OSCC)**, chosen because treatment side effects strongly affect speech, eating, social interaction, appearance, fatigue, and identity — domains that naturally emerge in conversational narratives.

The proposed system will:

- simulate synthetic patients from latent utility profiles,
- generate realistic interview responses,
- infer utilities from conversational transcripts,
- compare inferred utilities to known ground truth values, and
- evaluate the information value of different interview strategies.

The project will be implemented as an open-source, GitHub-hosted, API-free, browser-based research platform using GitHub Pages, client-side JavaScript, JSON-based experiment schemas, Quarto documentation, and human-mediated LLM interaction via copy/paste prompts.

2 Background and Motivation

2.1 Limitations of Conventional Utility Elicitation

Health preference elicitation methods frequently rely on abstract numerical or probabilistic reasoning tasks. While these methods are theoretically grounded, many patients find them cognitively difficult, emotionally unnatural, detached from lived experience, and difficult to interpret.

For example:

- Standard gamble requires reasoning about mortality probabilities.
- Time tradeoff requires hypothetical sacrifice of lifespan.
- Visual analog scales compress complex experiences into a single number.
- Multi-attribute utility methods may require explicit ranking of unfamiliar concepts.

These approaches may inadequately capture: identity disruption, social isolation, embarrassment, loss of meaningful activities, adaptation over time, and emotional salience of functional impairments.

2.2 Narrative Expression as Preference Evidence

Patients are generally much more comfortable describing:

- daily routines,
- frustrations,
- losses,
- fears,
- coping strategies,
- social experiences,
- changes in identity,
- disruptions to meaningful activities.

The central hypothesis of this project is:

> Patient utilities can be inferred from conversational narratives describing lived experience.

Rather than directly asking patients to numerically value health states, conversational elicitation attempts to infer latent preferences from:

- spontaneous topic salience,
- emotional framing,
- descriptions of daily functioning,
- tradeoff language,
- behavioral adaptations,
- social and emotional consequences.

3 Research Objectives

The primary objectives of the project are:

1. Develop a simulation environment for conversational utility elicitation.
2. Evaluate the recoverability of latent utility values from conversational transcripts.

3. Compare alternative elicitation question sets.
4. Investigate adaptive interviewing strategies.
5. Identify which utility domains are most recoverable conversationally.
6. Build an open-source benchmark platform for future research.

4 Oral Squamous Cell Carcinoma (OSCC) as the Initial Domain

4.1 Rationale for Selecting OSCC

OSCC is particularly suitable for conversational elicitation because treatment side effects strongly affect:

- speech,
- swallowing,
- eating,
- facial appearance,
- fatigue,
- social interaction,
- employment,
- intimacy,
- self-image.

These impacts naturally emerge in open-ended conversation.

For example:

- dysphagia may manifest as avoidance of restaurants,
- xerostomia may appear as descriptions of carrying water constantly,
- speech impairment may affect phone use or work,
- disfigurement may affect social participation and self-esteem.

4.2 Relationship to Existing OSCC Utility Models

This project will extend existing OSCC patient utility models by introducing:

- narrative simulation,
- conversational analysis,
- latent utility inference,
- round-trip recovery evaluation.

The project will incorporate:

- clinical states,

- symptom severity,
- treatment side effects,
- functional impairments,
- latent preference weights,
- adaptation behaviors,
- conversational response generation.

5 Conceptual Framework

5.1 Latent Utility Structure

Each synthetic patient will possess a latent utility vector describing the relative importance of health domains.

Example domains include:

Domain	Description
Speech function	Importance of verbal communication
Eating ability	Importance of unrestricted eating
Social participation	Importance of social engagement
Appearance	Sensitivity to facial disfigurement
Fatigue burden	Impact of reduced energy
Independence	Importance of self-sufficiency
Pain burden	Sensitivity to chronic pain
Employment role	Importance of work function
Family role	Importance of caregiving/social role

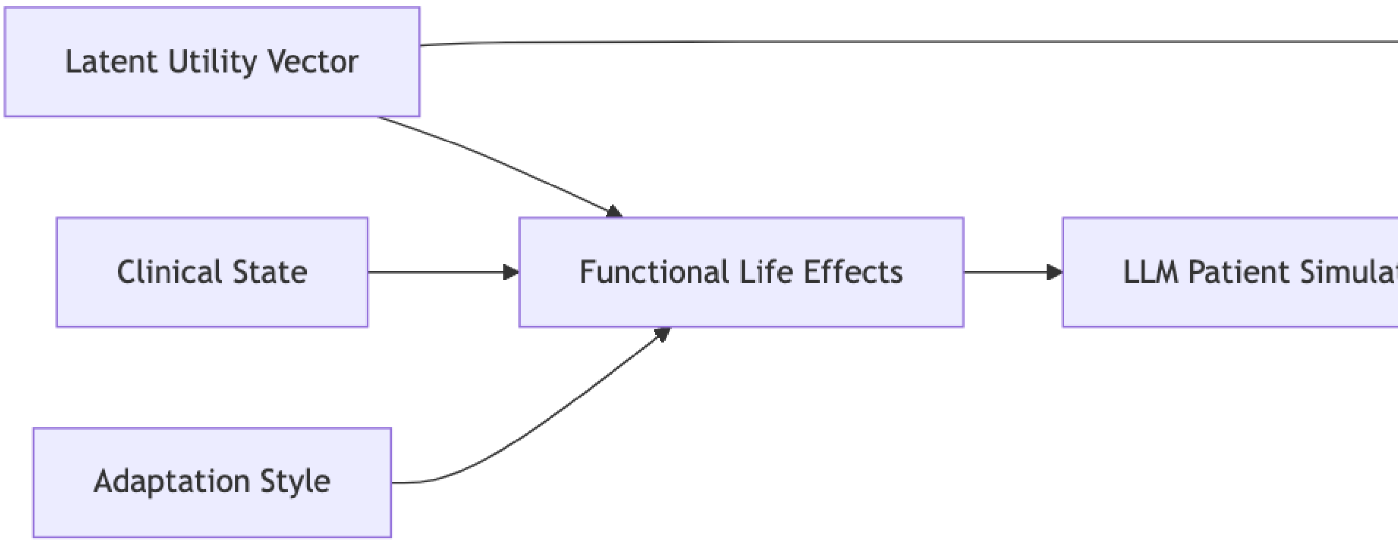
5.2 Multi-Layer Patient Representation

The project distinguishes several conceptually distinct layers.

Layer	Description
Clinical state	Objective disease/treatment status
Functional effects	Observable daily-life consequences
Utility structure	Latent patient priorities
Adaptation style	Psychological coping and normalization
Narrative expression	Conversational responses

6 System Architecture

6.1 High-Level Architecture



High-level conversational elicitation architecture

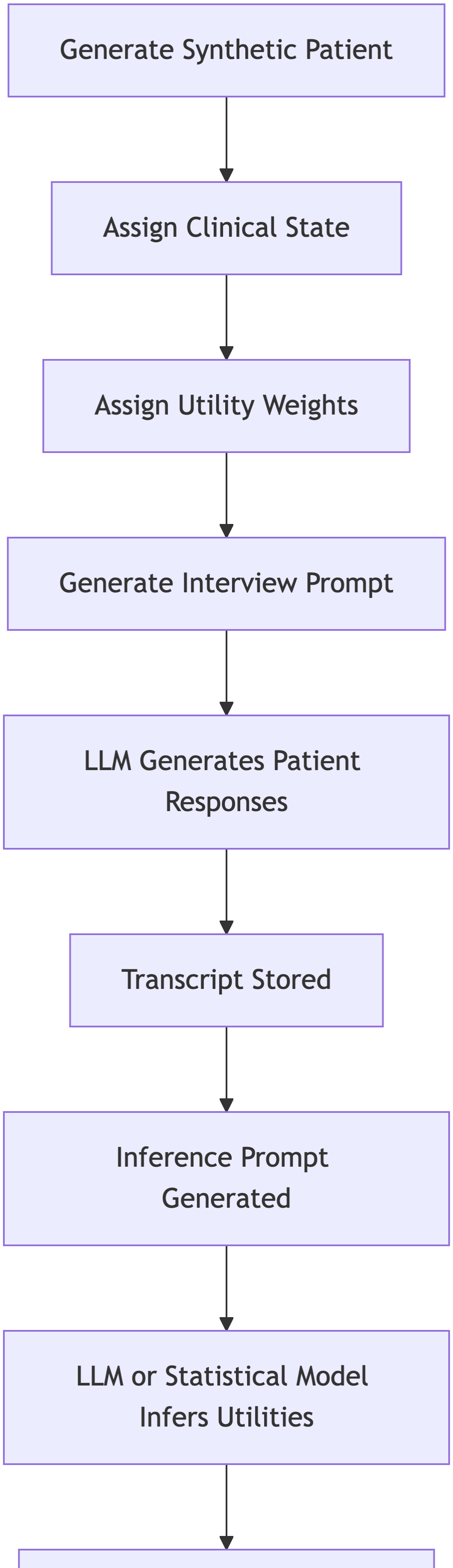
7 Round-Trip Simulation Framework

7.1 Overview

The project is built around a round-trip simulation process:

1. Generate latent utility profile.
2. Generate synthetic patient conversation.
3. Infer utilities from transcript.
4. Compare inferred and true utilities.

7.2 Round-Trip Workflow



Round-trip simulation workflow

8 Conversational Elicitation Strategy

8.1 Open-Ended Narrative Prompts

Examples: - “Tell me about your life.” - “What does a normal day look like?” - “What has changed since treatment?” - “What activities are most difficult now?”

These prompts encourage spontaneous emergence of: - salience, - identity disruption, - behavioral adaptation, - emotional priorities.

8.2 Functional Probes

Examples: - “What tasks take longer now?” - “How has eating changed?” - “How has treatment affected social interactions?”

These target specific functional domains.

8.3 Tradeoff-Oriented Prompts

Examples: - “If one aspect of your health improved, what would matter most?” - “What loss has been hardest to adapt to?”

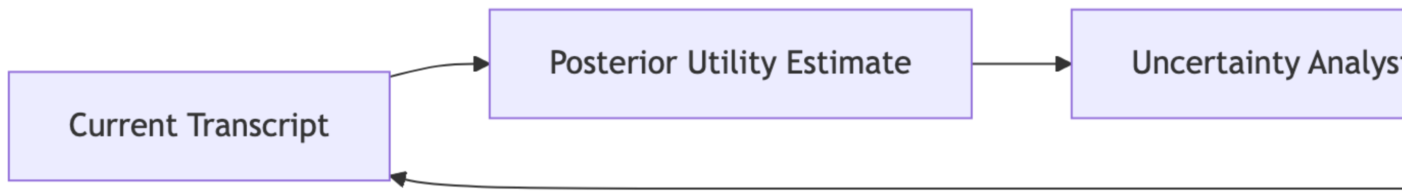
These may reveal relative utility weights more directly.

9 Adaptive Interviewing

The proposed framework supports adaptive conversational interviewing.

After each response: 1. the system estimates uncertainty in latent utilities, 2. identifies poorly constrained domains, 3. selects follow-up questions maximizing expected information gain.

9.1 Adaptive Interview Concept



Adaptive interviewing loop

10 Technical Architecture

10.1 Design Principles

The system will be: - open-source, - browser-based, - API-independent, - reproducible, - transparent, - low-cost, - forkable by researchers.

10.2 GitHub-Based Deployment

The project will be hosted on GitHub and deployed using GitHub Pages.

The application will: - run entirely client-side, - use static assets, - avoid server infrastructure, - avoid dependency on commercial APIs.

10.3 Human-Mediated LLM Interaction

Rather than using direct API calls: - prompts will be generated in-browser, - users will copy prompts into external LLM systems, - users will paste generated outputs back into the application.

Advantages include: - zero infrastructure cost, - model flexibility, - reproducibility, - transparency, - compatibility with local models, - avoidance of vendor lock-in.

11 Software Components

11.1 Proposed Repository Structure

```
conversational-elicitation/
  index.html
  data/
     oscc_attributes.json
     question_sets.json
     synthetic_patients.json
     prompt_templates.json
  js/
     app.js
     patient_generator.js
     prompt_builder.js
     transcript_parser.js
     utility_inference.js
     metrics.js
     visualization.js
  examples/
  docs/
  schema/
```

12 Data Model

12.1 Example Experiment Object

```json
{
  "experiment_id": "oscc_rt_001",
  "clinical_state": {
    "xerostomia": 0.8,
    "dysphagia": 0.6,
    "speech_impairment": 0.7
  },
  "utility_weights": {
    "social_function": 0.9,
    "appearance": 0.4,
    "independence": 0.8
  },
  "adaptation_style": {
    "stoicism": 0.7
  },
  "question_set": [],
```

```
  "transcript": "",
  "inferred_utilities": {},
  "metrics": {}
}
```

13 Utility Inference Approaches

13.1 LLM-Based Inference

An LLM may be prompted to: - identify salient themes, - estimate latent utility weights, - summarize burdens, - identify tradeoffs.

This provides a rapid baseline approach.

13.2 Embedding-Based Regression

Potential future methods: - transformer embeddings, - Bayesian regression, - Gaussian processes, - metric learning, - latent variable models.

13.3 Bayesian Inference Models

Longer-term development may incorporate: - probabilistic graphical models, - posterior uncertainty estimation, - latent narrative-generation models, - active learning strategies.

14 Evaluation Metrics

14.1 Recovery Accuracy

Potential metrics include: - root mean squared error, - rank correlation, - calibration error, - posterior entropy, - confidence interval coverage.

14.2 Identifiability Analysis

The project will investigate: - which utility dimensions are recoverable, - which require direct probing, - ambiguity between utility domains, - information value of interview questions.

15 Ethical Considerations

15.1 Synthetic Patients vs Real Patients

Initial development will rely exclusively on synthetic patients.

No clinical deployment is proposed in the initial phase.

15.2 Risks

Potential risks include: - demographic stereotyping, - over-interpretation of narratives, - privacy concerns, - inappropriate automation of preference assessment.

15.3 Human Oversight

Conversational elicitation should be viewed as decision support, exploratory preference assessment, and augmentation of clinician understanding, not a replacement for clinical judgment.

16 Proposed Development Phases

16.1 Phase 1 — MVP Prototype

Features:

- OSCC utility schema,
- synthetic patient generation,
- transcript storage,
- prompt generation,
- utility comparison dashboard.

16.2 Phase 2 — Expanded Simulation

Features:

- multiple patient archetypes,
- adaptation styles,
- larger question libraries,
- batch simulation.

16.3 Phase 3 — Adaptive Interviewing

Features:

- uncertainty estimation,
- active question selection,
- posterior updating.

16.4 Phase 4 — Real-World Validation

Potential future activities:

- pilot clinician studies,
- comparison against standard utility instruments,
- qualitative usability studies.

17 Expected Contributions

This project may contribute:

- a new framework for conversational utility elicitation,
- a benchmark environment for preference inference,
- synthetic patient corpora,
- adaptive interviewing methodologies,
- narrative-based utility modeling approaches,
- open-source research infrastructure.

18 Conclusion

This proposal describes a novel framework for estimating patient utilities through conversational narratives rather than direct numerical elicitation.

The central innovation is a round-trip simulation architecture in which:

- latent utilities generate synthetic patient narratives,
- conversational transcripts are analyzed to recover utilities,
- recovery accuracy is quantitatively evaluated.

The project combines:

- health preference research,

- narrative medicine,
- probabilistic inference,
- conversational AI,
- simulation-based evaluation,
- open-source software engineering.

The proposed GitHub-hosted, API-free architecture emphasizes transparency, reproducibility, accessibility, extensibility, and collaborative experimentation.

OSCC provides a clinically meaningful and conversationally rich initial domain for evaluating whether conversational elicitation can recover latent patient preferences with sufficient fidelity to support future healthcare decision-making research.